

Digital IC Design

Project 1

“Spartan6 - DSP48A1”

Submitted by

Name / Hatem Essam-Eldein Amin

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Question 1

1. RTL Code

```
// =====  
// Generic Parametric Pipeline Register Module  
// =====  
module Pipeline_Module #(  
    parameter INPUT_SIZE = 8,  
    parameter RSTTYPE     = "SYNC", // "SYNC" or "ASYNC"  
    parameter PIPELINE    = 1       // 1 = Registered, 0 = Direct Bypass  
)(  
    input  wire [INPUT_SIZE-1:0] in_port,  
    input  wire                  clk,  
    input  wire                  rst,  
    input  wire                  ce,  
    output wire [INPUT_SIZE-1:0] out_Port  
);  
  
    reg [INPUT_SIZE-1:0] reg_out;  
  
    generate  
        if (RSTTYPE == "ASYNC") begin : g_async  
            always @(posedge clk or posedge rst) begin  
                if (rst) begin  
                    reg_out <= {INPUT_SIZE{1'b0}};  
                end else if (ce) begin  
                    reg_out <= in_port;  
                end  
            end  
        end else begin : g_sync  
            always @(posedge clk) begin  
                if (rst) begin  
                    reg_out <= {INPUT_SIZE{1'b0}};  
                end else if (ce) begin  
                    reg_out <= in_port;  
                end  
            end  
        end  
    endgenerate  
  
    // Selection mux: output registered value if PIPELINE is enabled, else bypass  
    assign out_Port = (PIPELINE) ? reg_out : in_port;  
  
endmodule  
  
// =====  
// Top-Level DSP48A1 Module  
// =====  
module DSP48A1 #(  
    parameter A0REG      = 0,  
    parameter A1REG      = 1,
```

```

parameter B0REG      = 0,
parameter B1REG      = 1,
parameter CREG       = 1,
parameter DREG       = 1,
parameter MREG       = 1,
parameter PREG       = 1,
parameter CARRYINREG = 1,
parameter CARRYOUTREG = 1,
parameter OPMODEREG  = 1,
parameter CARRYINSEL = "OPMODE5", // "OPMODE5" or "CARRYIN"
parameter B_INPUT    = "DIRECT",  // "DIRECT" or "CASCADE"
parameter RSTTYPE     = "SYNC"    // "SYNC" or "ASYN"
)(
    input wire [17:0] A,
    input wire [17:0] B,
    input wire [17:0] D,
    input wire [47:0] C,
    input wire        CLK,
    input wire        CARRYIN,
    input wire [7:0]  OPMODE,
    input wire [17:0] BCIN,
    input wire [47:0] PCIN,

    input wire RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE,
    input wire CEA,  CEB,  CEM,  CEP,  CEC,  CED,  CECARRYIN,  CEOPMODE,

    output wire [17:0] BCOUT,
    output wire [47:0] PCOUT,
    output wire [47:0] P,
    output wire [35:0] M,
    output wire        CARRYOUT,
    output wire        CARRYOUTF
);

// -----
// Stage 1 Pipeline Registers: D, B0, A0, C, OPMODE, CARRYIN
// -----
wire [17:0] b_src = (B_INPUT == "CASCADE") ? BCIN : B;

wire [17:0] d0;
wire [17:0] b0;
wire [17:0] a0;
wire [47:0] c0;
wire [7:0]  opmode0;
wire        cyi0;

Pipeline_Module #(.INPUT_SIZE(18), .RSTTYPE(RSTTYPE), .PIPELINE(DREG))
    D_REG (.in_port(D), .out_Port(d0), .clk(CLK), .rst(RSTD), .ce(CED));

Pipeline_Module #(.INPUT_SIZE(18), .RSTTYPE(RSTTYPE), .PIPELINE(B0REG))
    B0_REG (.in_port(b_src), .out_Port(b0), .clk(CLK), .rst(RSTB), .ce(CEB));

Pipeline_Module #(.INPUT_SIZE(18), .RSTTYPE(RSTTYPE), .PIPELINE(A0REG))
    A0_REG (.in_port(A), .out_Port(a0), .clk(CLK), .rst(RSTA), .ce(CEA));

```

```

Pipeline_Module #(.INPUT_SIZE(48), .RSTTYPE(RSTTYPE), .PIPELINE(CREG))
    C_REG (.in_port(C), .out_Port(c0), .clk(CLK), .rst(RSTC), .ce(CEC));

Pipeline_Module #(.INPUT_SIZE(8), .RSTTYPE(RSTTYPE), .PIPELINE(OPMODEREG))
    OPMODE_REG (.in_port(OPMODE), .out_Port(opmode0), .clk(CLK), .rst(RSTOPMODE),
.ce(CEOPMODE));

Pipeline_Module #(.INPUT_SIZE(1), .RSTTYPE(RSTTYPE), .PIPELINE(CARRYINREG))
    CYI_REG (.in_port(CARRYIN), .out_Port(cyi0), .clk(CLK), .rst(RSTCARRYIN),
.ce(CECARRYIN));

// -----
// Pre-adder / Pre-subtractor Stage (D +/- B)
// -----
wire [17:0] preadder_out = opmode0[6] ? (d0 - b0) : (d0 + b0);
wire [17:0] b1_in      = opmode0[4] ? preadder_out : b0;

// -----
// Stage 2 Pipeline Registers: A1, B1
// -----
wire [17:0] a1;
wire [17:0] b1;

Pipeline_Module #(.INPUT_SIZE(18), .RSTTYPE(RSTTYPE), .PIPELINE(A1REG))
    A1_REG (.in_port(a0), .out_Port(a1), .clk(CLK), .rst(RSTA), .ce(CEA));

Pipeline_Module #(.INPUT_SIZE(18), .RSTTYPE(RSTTYPE), .PIPELINE(B1REG))
    B1_REG (.in_port(b1_in), .out_Port(b1), .clk(CLK), .rst(RSTB), .ce(CEB));

assign BCOUT = b1;

// -----
// Multiplier & Concatenation (D:A:B) Stage
// -----
wire signed [35:0] mult_comb = $signed(a1) * $signed(b1);
wire [47:0] dab_comb = {d0[11:0], a1, b1};

wire [35:0] m_val;
wire [47:0] dab_val;

Pipeline_Module #(.INPUT_SIZE(36), .RSTTYPE(RSTTYPE), .PIPELINE(MREG))
    M_REG (.in_port(mult_comb), .out_Port(m_val), .clk(CLK), .rst(RSTM), .ce(CEM));

Pipeline_Module #(.INPUT_SIZE(48), .RSTTYPE(RSTTYPE), .PIPELINE(MREG))
    DAB_REG (.in_port(dab_comb), .out_Port(dab_val), .clk(CLK), .rst(RSTM), .ce(CEM));

assign M = m_val;
wire [47:0] mult_ext = {{12{m_val[35]}}}, m_val};

// -----
// Carry-In Mux & Post-Adder Operand Multiplexers (X and Z)
// -----
wire cin_final = (CARRYINSEL == "CARRYIN") ? cyi0 : opmode0[5];

```

```

wire [47:0] p_val; // Feedback from final P register output

wire [47:0] x_mux = (opmode0[1:0] == 2'b00) ? 48'd0      :
                    (opmode0[1:0] == 2'b01) ? mult_ext :
                    (opmode0[1:0] == 2'b10) ? p_val      :dab_val;

wire [47:0] z_mux = (opmode0[3:2] == 2'b00) ? 48'd0 :
                    (opmode0[3:2] == 2'b01) ? PCIN  :
                    (opmode0[3:2] == 2'b10) ? p_val :c0;

wire x_is_zero    = (opmode0[1:0] == 2'b00);
wire z_is_zero    = (opmode0[3:2] == 2'b00);
wire bypass_adder = x_is_zero || z_is_zero;

// -----
// Post-Adder / Subtractor Execution Logic
// -----
wire [48:0] add_ext = {1'b0, x_mux} + {1'b0, z_mux} + cin_final;
wire [48:0] sub_ext = {1'b0, z_mux} - {1'b0, x_mux} - cin_final;

wire [47:0] p_comb = bypass_adder ? (z_is_zero ? x_mux : z_mux)
                                   : (opmode0[7] ? sub_ext[47:0] : add_ext[47:0]);

wire carry_comb = opmode0[7] ? sub_ext[48] : add_ext[48];

// -----
// Output Registers: Carry Out & P
// -----
wire carryout_val;

// Feed carry_comb or freeze current carryout_val based on bypass status
wire carry_in_sel = bypass_adder ? carryout_val : carry_comb;

Pipeline_Module #(.INPUT_SIZE(1), .RSTTYPE(RSTTYPE), .PIPELINE(CARRYOUTREG))
  CYO_REG (.in_port(carry_in_sel), .out_Port(carryout_val), .clk(CLK),
    .rst(RSTCARRYIN), .ce(CECARRYIN));

Pipeline_Module #(.INPUT_SIZE(48), .RSTTYPE(RSTTYPE), .PIPELINE(PREG))
  P_REG (.in_port(p_comb), .out_Port(p_val), .clk(CLK), .rst(RSTP), .ce(CEP));

// Final Output Assignments
assign P      = p_val;
assign PCOUT  = P;
assign CARRYOUT = carryout_val;
assign CARRYOUTF= CARRYOUT;

endmodule

```

2. Testbench code

```
`timescale 1ns / 1ps

module DSP48A1_tb;

    // -----
    // Testbench Parameters & Signals
    // -----
    reg [17:0] A, B, D, BCIN;
    reg [47:0] C, PCIN;
    reg [7:0] OPMODE;
    reg      CLK, CARRYIN;

    reg      RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE;
    reg      CEA, CEB, CEM, CEP, CEC, CED, CECARRYIN, CEOPMODE;

    wire [17:0] BCOUT;
    wire [47:0] PCOUT, P;
    wire [35:0] M;
    wire      CARRYOUT, CARRYOUTF;

    integer    error_count = 0;
    reg [47:0] past_P;
    reg      past_CARRYOUT;

    // -----
    // DUT Instantiation (Default Parameter Configuration)
    // -----
    DSP48A1 #(
        .A0REG(0),
        .A1REG(1),
        .B0REG(0),
        .B1REG(1),
        .CREG(1),
        .DREG(1),
        .MREG(1),
        .PREG(1),
        .CARRYINREG(1),
        .CARRYOUTREG(1),
        .OPMODEREG(1),
        .CARRYINSEL("OPMODE5"),
        .B_INPUT("DIRECT"),
        .RSTTYPE("SYNC")
    ) DUT (
        .A(A), .B(B), .D(D), .C(C),
        .CLK(CLK), .CARRYIN(CARRYIN), .OPMODE(OPMODE),
        .BCIN(BCIN), .PCIN(PCIN),
        .RSTA(RSTA), .RSTB(RSTB), .RSTM(RSTM), .RSTP(RSTP),
        .RSTC(RSTC), .RSTD(RSTD), .RSTCARRYIN(RSTCARRYIN), .RSTOPMODE(RSTOPMODE),
        .CEA(CEA), .CEB(CEB), .CEM(CEM), .CEP(CEP),
        .CEC(CEC), .CED(CED), .CECARRYIN(CECARRYIN), .CEOPMODE(CEOPMODE),
        .BCOUT(BCOUT), .PCOUT(PCOUT), .P(P), .M(M),
```

```

        .CARRYOUT(CARRYOUT), .CARRYOUTF(CARRYOUTF)
    );

    // -----
    // Clock Generation (10ns Period)
    // -----
    initial begin
        CLK = 0;
        forever #5 CLK = ~CLK;
    end

    // -----
    // Self-Checking Helper Task
    // -----
    task check_outputs;
        input [17:0] exp_BCOUNT;
        input [35:0] exp_M;
        input [47:0] exp_P;
        input      exp_CARRYOUT;
        input [8*20:1] test_name;
        begin
            if (BCOUNT === exp_BCOUNT && M === exp_M && P === exp_P &&
                PCOUNT === exp_P && CARRYOUT === exp_CARRYOUT && CARRYOUTF ===
exp_CARRYOUT) begin
                $display("[PASS] %s: BCOUNT='h%h", M='h%h", P=PCOUNT='h%h", CARRYOUT='h%b",
                        test_name, BCOUNT, M, P, CARRYOUT);
            end else begin
                $display("[FAIL] %s:", test_name);
                if (BCOUNT !== exp_BCOUNT) $display("    -> BCOUNT: Expected 'h%h", Got
'h%h", exp_BCOUNT, BCOUNT);
                if (M !== exp_M) $display("    -> M      : Expected 'h%h", Got
'h%h", exp_M, M);
                if (P !== exp_P) $display("    -> P      : Expected 'h%h", Got
'h%h", exp_P, P);
                if (PCOUNT !== exp_P) $display("    -> PCOUNT: Expected 'h%h", Got
'h%h", exp_P, PCOUNT);
                if (CARRYOUT !== exp_CARRYOUT)$display("    -> CYOUT: Expected 'h%b", Got
'h%b", exp_CARRYOUT, CARRYOUT);
                error_count = error_count + 1;
            end
        end
    endtask

    // -----
    // Stimulus & Verification Sequence
    // -----
    initial begin
        $display("=====");
        $display("      Starting DSP48A1 Verification      ");
        $display("=====");

        // -----
        // Step 2.1: Verify Reset Operation
        // -----

```

```

RSTA = 1; RSTB = 1; RSTM = 1; RSTP = 1;
RSTC = 1; RSTD = 1; RSTCARRYIN = 1; RSTOPMODE = 1;

CEA = 0; CEB = 0; CEM = 0; CEP = 0;
CEC = 0; CED = 0; CECARRYIN = 0; CEOPMODE = 0;

// Drive remaining inputs with random values
A = $urandom; B = $urandom; D = $urandom; C = {$urandom, $urandom};
BCIN = $urandom; PCIN = {$urandom, $urandom};
OPMODE = $urandom; CARRYIN = $urandom % 2;

@(negedge CLK);
check_outputs(18'h0, 36'h0, 48'h0, 1'b0, "2.1 Reset Test");

// Deassert resets & enable clock enables
RSTA = 0; RSTB = 0; RSTM = 0; RSTP = 0;
RSTC = 0; RSTD = 0; RSTCARRYIN = 0; RSTOPMODE = 0;

CEA = 1; CEB = 1; CEM = 1; CEP = 1;
CEC = 1; CED = 1; CECARRYIN = 1; CEOPMODE = 1;

// -----
// Step 2.2: Verify DSP Path 1
// -----
OPMODE = 8'b11011101;
A = 20;
B = 10;
C = 350;
D = 25;
BCIN = $urandom;
PCIN = {$urandom, $urandom};
CARRYIN = $urandom % 2;

repeat(4) @(negedge CLK);
check_outputs(18'hF, 36'h12C, 48'h32, 1'b0, "2.2 DSP Path 1");

// -----
// Step 2.3: Verify DSP Path 2
// -----
OPMODE = 8'b00010000;
A = 20;
B = 10;
C = 350;
D = 25;
BCIN = $urandom;
PCIN = {$urandom, $urandom};
CARRYIN = $urandom % 2;

repeat(3) @(negedge CLK);
check_outputs(18'h23, 36'h2BC, 48'h0, 1'b0, "2.3 DSP Path 2");

// -----
// Step 2.4: Verify DSP Path 3
// -----

```

```
// Capture prior P and CARRYOUT values for self-checking dynamic comparison
past_P      = P;
past_CARRYOUT = CARRYOUT;
```

```
OPMODE = 8'b00001010;
A       = 20;
B       = 10;
C       = 350;
D       = 25;
BCIN    = $urandom;
PCIN    = {$urandom, $urandom};
CARRYIN = $urandom % 2;
```

```
repeat(3) @(negedge CLK);
check_outputs(18'hA, 36'hC8, past_P, past_CARRYOUT, "2.4 DSP Path 3");
```

```
// -----
// Step 2.5: Verify DSP Path 4
// -----
```

```
OPMODE = 8'b10100111;
A       = 5;
B       = 6;
C       = 350;
D       = 25;
PCIN    = 3000;
BCIN    = $urandom;
CARRYIN = $urandom % 2;
```

```
repeat(3) @(negedge CLK);
check_outputs(18'h6, 36'h1E, 48'hFE6FFFE0BB1, 1'b1, "2.5 DSP Path 4");
```

```
// -----
// End of Simulation Summary
// -----
```

```
$display("=====");
if (error_count == 0)
    $display("  TESTBENCH SUCCESSFUL: All tests passed!");
else
    $display("  TESTBENCH FAILED: %0d error(s) detected.", error_count);
$display("=====");
```

```
$finish;
```

```
end
```

```
endmodule
```

3. Do file

vlib work

vlog DSP48A1.v DSP48A1_tb.v

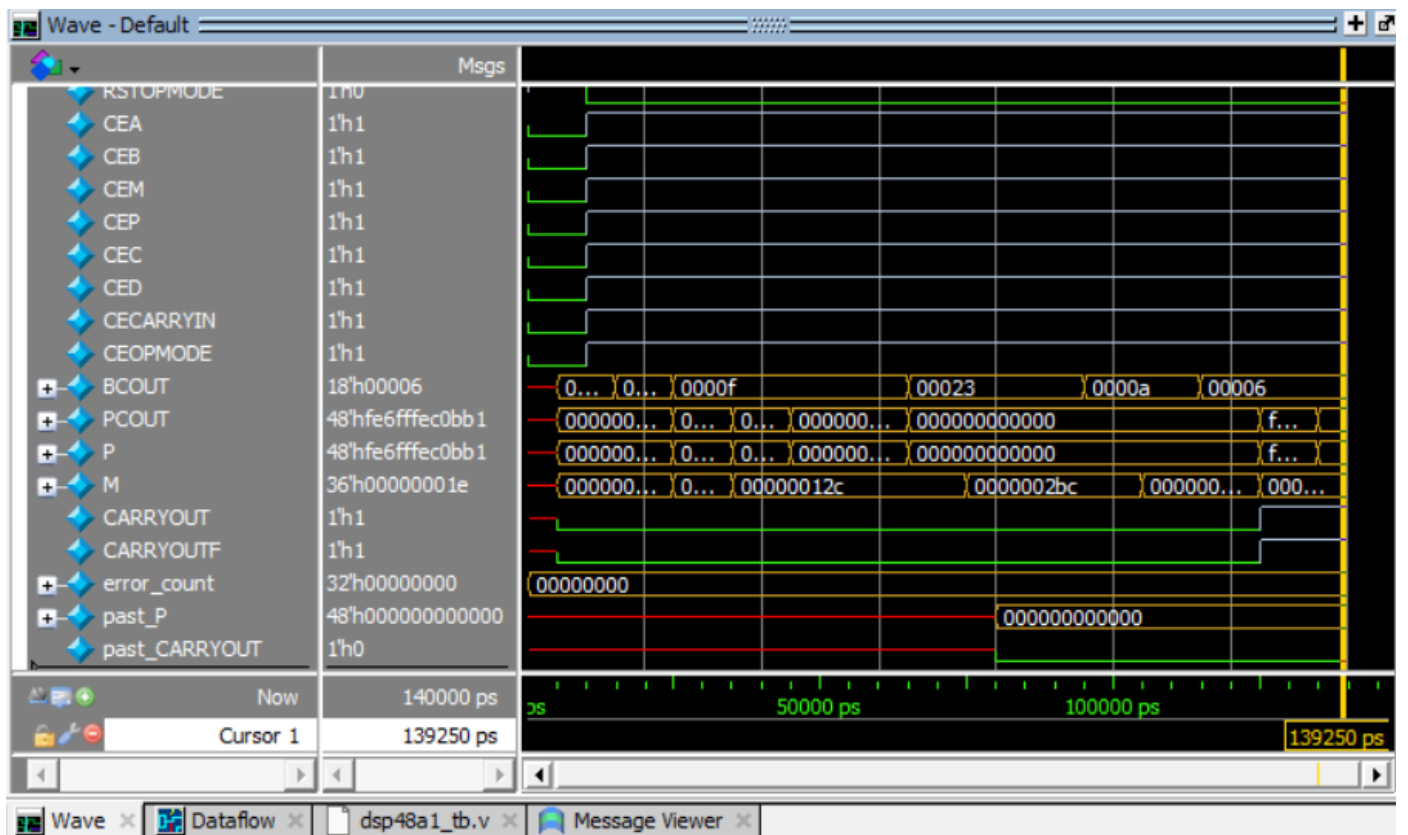
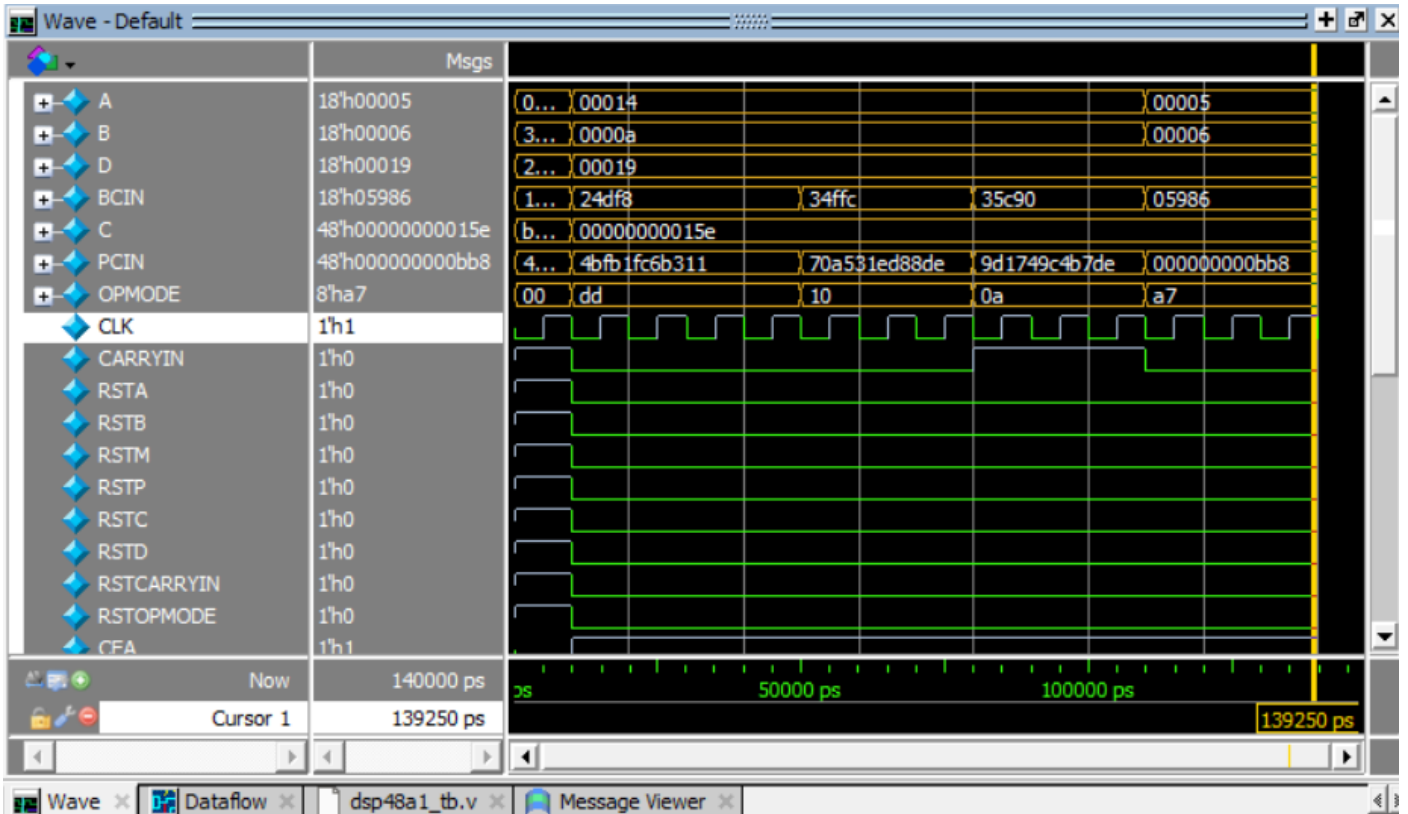
vsim -voptargs=+acc work.DSP48A1_tb

add wave *

run -all

#quit -sim

4. QuestaSim Snippets



5. Constraint File

```
6. ## Clock constraint: 100 MHz on pin W5 (xc7a200tffg1156-3)
7. set_property PACKAGE_PIN W5 [get_ports CLK]
8. set_property IOSTANDARD LVCMOS33 [get_ports CLK]
9. create_clock -period 10.000 -name sys_clk_pin -waveform {0.000 5.000} [get_ports
  CLK]
10.
```

11. Elaboration (“Messages” tab & Schematic snippets)

ELABORATED DESIGN - xc7a200tffg1156-3 (active)

Sources Netlist x ? _ □ ×

DSP48A1

- > Nets (1197)
- > Leaf Cells (24)
- > A0_REG (Pipeline_Module__parameterized0)
- > A1_REG (Pipeline_Module)
- > B0_REG (Pipeline_Module__parameterized0)
- > B1_REG (Pipeline_Module)
- > C_REG (Pipeline_Module__parameterized1)

Properties ? _ □ ×

Select an object to see properties

Project Summary x Schematic x ? _ □ ×

35 Cells 346 I/O Ports 1197 Nets

Tcl Console Messages x Log Reports Design Runs ? _ □ ×

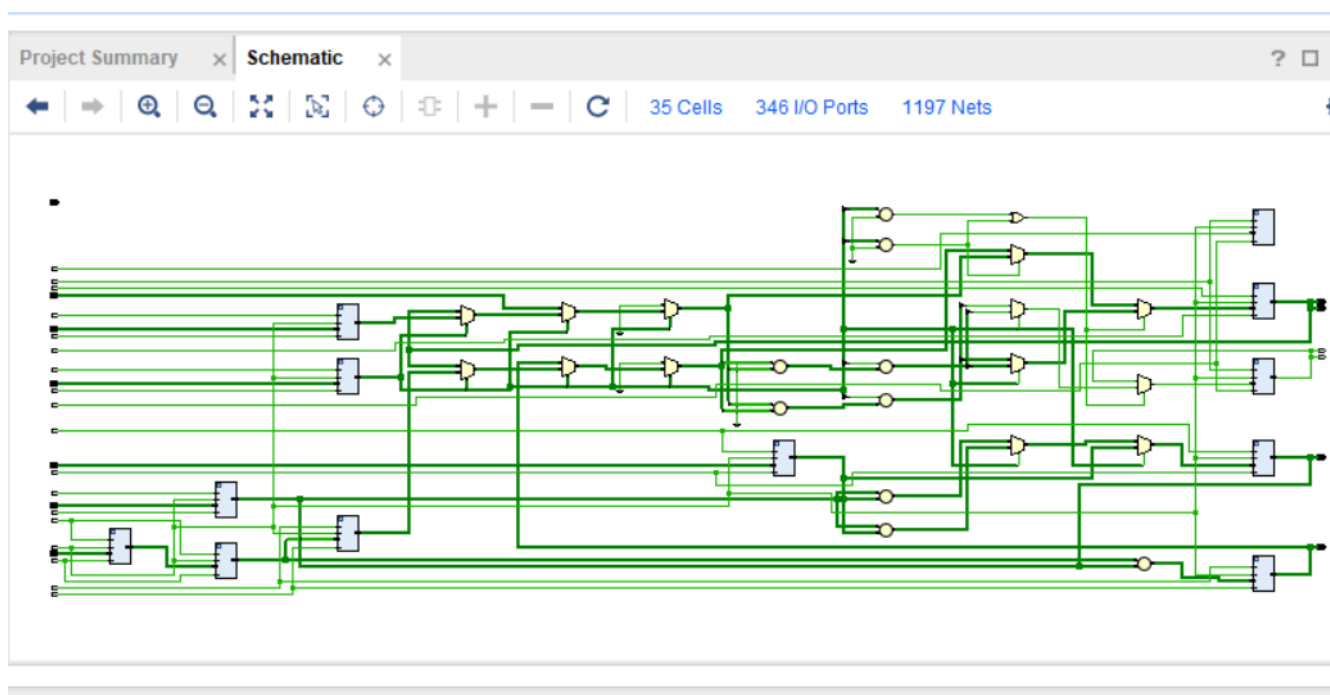
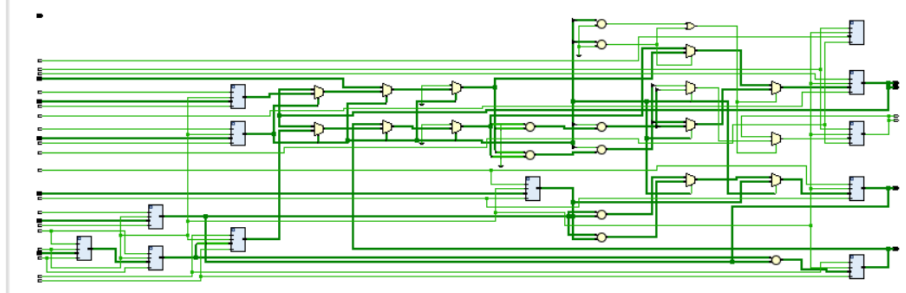
Warning (22) Info (21) Status (11) Show All

Vivado Commands (3 infos)

- General Messages (3 infos)
- [IP_Flow 19-234] Refreshing IP repositories
- [IP_Flow 19-1704] No user IP repositories specified
- [IP_Flow 19-2313] Loaded Vivado IP repository 'C:/Xilinx/Vivado/2018.2/data/ip'

Elaborated Design (22 warnings, 18 infos)

- General Messages (22 warnings, 18 infos)
- [Synth 8-2490] overwriting previous definition of module DSP48A1 [dsp48a1.v:47]
- [Synth 8-6157] synthesizing module 'DSP48A1' [dsp48a1.v:47] (6 more like this)



12. Synthesis (“Messages” tab, Utilization report, timing report & Schematic snippets)

Flow Navigator

- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Open Elaborated Design
 - Report Methodology
 - Report DRC
 - Report Noise
 - Schematic
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
 - Constraints Wizard
 - Edit Timing Constraints
 - Set Up Debug
 - Report Timing Summary
 - Report Clock Networks
 - Report Clock Interaction
 - Report Methodology

SYNTHESIZED DESIGN - xc7a200tfg1156-3 (active)

Sources Netlist

- C (48)
- C_IBUF (48)
- D (18)
- D_IBUF (18)
- M (36)
- M_OBUF (36)
- M_OBUF[0]
- M_OBUF[1]

Bus Net Properties

Name: C

General Scalar Nets

Schematic

337 Cells 346 I/O Ports 825 Nets

Net: RSTCARRYIN
Type: SIGNAL
Route status: Has unplaced ports or pins.

Tcl Console Messages Log Reports Design Runs Debug

Warning (65) Info (57) Status (28) Show All

Vivado Commands (3 infos)

- General Messages (3 infos)
 - [IP_Flow 19-234] Refreshing IP repositories
 - [IP_Flow 19-1704] No user IP repositories specified
 - [IP_Flow 19-2313] Loaded Vivado IP repository 'C:/xilinx/Vivado/2018.2/data/ip'
- Elaborated Design (22 warnings, 18 infos)
 - General Messages (22 warnings, 18 infos)
 - [Synth 8-2490] overwriting previous definition of module DSP48A1 [dsp48a1.v47]
 - [Synth 8-5157] synthesizing module 'DSP48A1' [dsp48a1.v47] (6 more like this)

Net: RSTCARRYIN Type: SIGNAL Route status: Has unplaced ports or pins

SYNTHESIZED DESIGN - xc7a200tfg1156-3 (active)

Sources

Netlist

Bus Net Properties

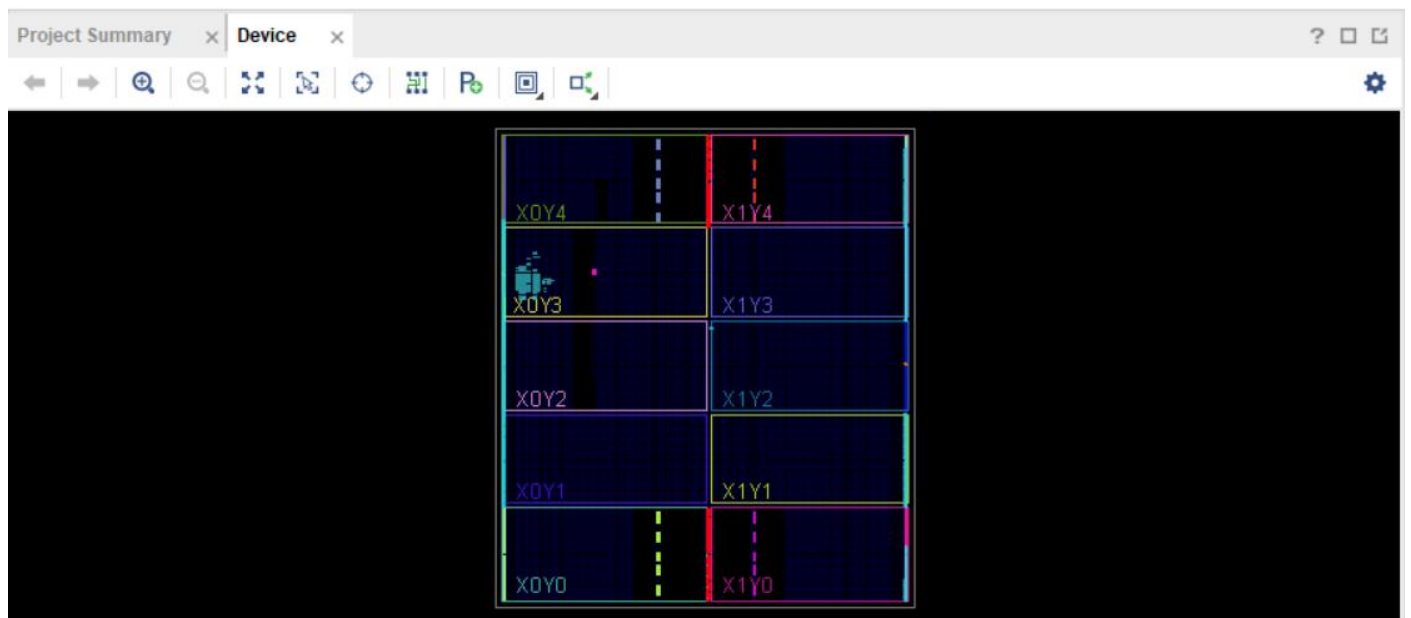
Schematic

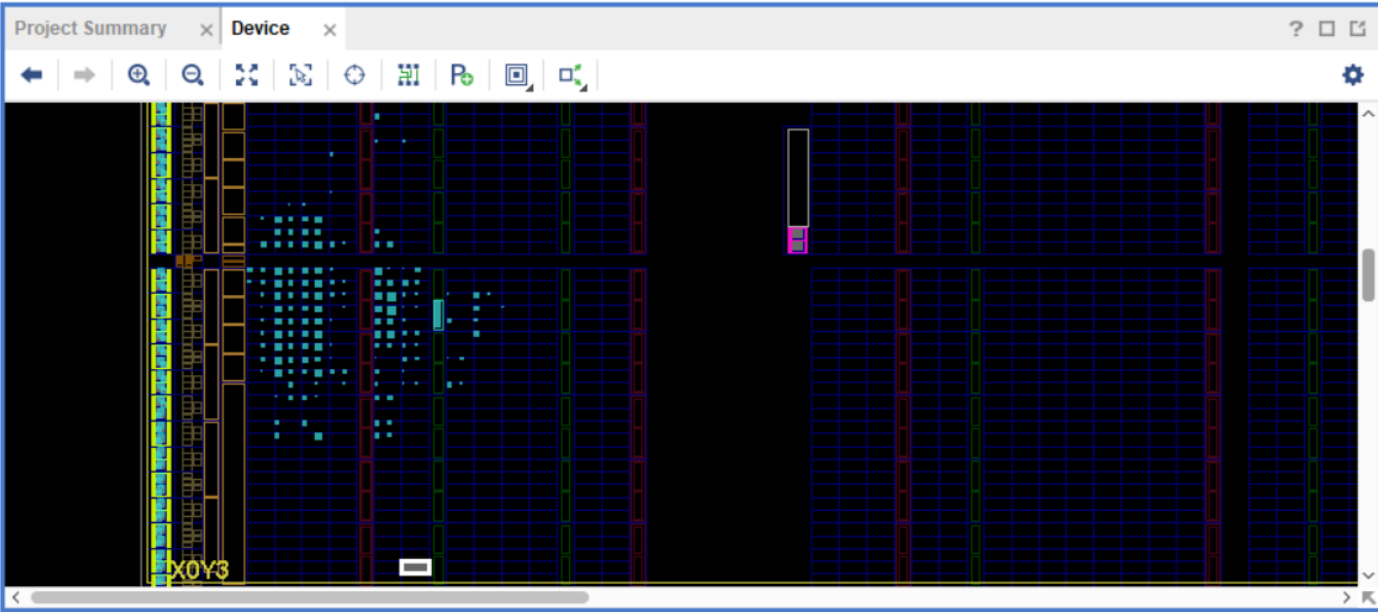
337 Cells 346 I/O Ports 825 Nets

Tcl Console Messages Log Reports Design Runs Utilization x Timing Debug						
Hierarchy						
Name	Slice LUTs (134600)	Slice Registers (269200)	DSPs (740)	Bonded IOB (500)	BUFGCTRL (32)	
DSP48A1	304	207	1	327	1	
A1_REG (Pipeline_Mod...)	0	18	0	0	0	
B1_REG (Pipeline_Mod...)	0	18	0	0	0	
C_REG (Pipeline_Mod...)	0	48	0	0	0	

Tcl Console Messages Log Reports Design Runs Timing x Debug			
Design Timing Summary			
Setup	Hold	Pulse Width	
Worst Negative Slack (WNS): 5.430 ns	Worst Hold Slack (WHS): 0.140 ns	Worst Pulse Width Slack (WPWS): 4.500 ns	
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns	
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	
Total Number of Endpoints: 133	Total Number of Endpoints: 133	Total Number of Endpoints: 209	
All user specified timing constraints are met.			

13. Implementation (“Messages” tab, Utilization report, timing report & device snippets)





Tcl Console Messages Log Reports Design Runs Power DRC Methodology Timing Utilization x

Hierarchy

Name	Slice LUTs (133800)	Slice Registers (267600)	Slice (33450)	LUT as Logic (133800)	LUT Flip Flop Pairs (133800)	DSPs (740)	Bonded IOB (500)	BUFGCTRL (32)
▼ DSP48A1	303	225	148	303	49	1	327	1
A1_REG (Pipeline_Mo...	0	18	7	0	0	0	0	0
B1_REG (Pipeline_Mo...	0	36	16	0	0	0	0	0
C_REG (Pipeline_Mod...	0	48	16	0	0	0	0	0

utilization_1

Tcl Console Messages Log Reports Design Runs Power DRC Methodology Timing x

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.316 ns	Worst Hold Slack (WHS): 0.142 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 151	Total Number of Endpoints: 151	Total Number of Endpoints: 227

All user specified timing constraints are met.

Timing Summary - impl_1 (saved)

14. Linting (snippets showing no errors)

Questa Lint 2021.1 (...rojects/Project_1/Project1_Final/lint.db)

File Edit View Schematic Window Help

Design

Search: Type Search...

Instance Module

DSP48A1 (12) DSP48A1

```
43 // =====
44 // Top-Level DSP48A1 Module
45 // =====
46 module DSP48A1 #(
47     parameter A0REG = 0,
48     parameter A1REG = 1,
49     parameter B0REG = 0,
50     parameter B1REG = 1,
51     parameter CREG = 1,
52     parameter DREG = 1,
53     parameter MREG = 1,
54     parameter PREG = 1,
55     parameter CARRYINREG = 1,
56     parameter CARRYOUTREG = 1,
57     1
58 )
```

Schematic 1

Flow Navigator Design

Lint Checks

Filter: Type here

Waived Fixed Pending Uninspected Bug Verified Total : 5 Selected : 0

Severity	Status	Check	Alias	Message	Module	Category	State	Owner	STARC Reference
Warning	2	condition_const		Condition expression is a constant. Module Pipeline_...	Pipeline_Module	Rtl Design Style	open	unassign...	
Warning	2	condition_const		Condition expression is a constant. Module DSP48A1,...	DSP48A1	Rtl Design Style	open	unassign...	
Warning	2	condition_const		Condition expression is a constant. Module DSP48A1,...	DSP48A1	Rtl Design Style	open	unassign...	
Warning	2	parameter_name_duplicate		Same parameter name is used in more than one mod...	DSP48A1	Nomenclature...	open	unassign...	1.1.4.3, 3.2.2.2
Warning	2	multi_ports_in_single_line		Multiple ports are declared in one line. Module DSP48...	DSP48A1	Rtl Design Style	open	unassign...	3.5.6.3

Transcript Message Viewer Lint Checks Design Metrics Design Information Status History Lint Dashboard

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